

The Generative AI Revolution

From Discriminative to Generative AI

Traditional (discriminative) AI models classify or predict from existing data: 'Is this email spam?', 'What digit is in this image?'. They map inputs to labels.

Generative AI goes further — it creates new content: images, text, audio, code, and video that did not previously exist. It learns the underlying distribution of data and can sample from it.

Discriminative AI	Generative AI
Classifies / predicts	Creates / generates
Maps input → label	Samples new data from learned distribution
Example: spam filter	Example: ChatGPT, DALL-E, Suno
Output: a category or number	Output: text, image, audio, video

The Three Core Shifts of the Generative AI Revolution

Shift 1 — Scale: Models grew from millions to hundreds of billions of parameters. GPT-3 (175B), GPT-4 (est. 1T+). Scale alone unlocked emergent capabilities.

Shift 2 — Multimodality: Models moved beyond text to handle images, audio, video, and code within a single architecture (e.g., GPT-4o, Gemini).

Shift 3 — Accessibility: API-first deployment democratised AI — any developer can now integrate world-class AI with a few lines of code, no ML expertise required.

The generative AI revolution is not just a technical upgrade — it is a platform shift comparable to the internet or the smartphone.

Real-World Generative AI Applications

- Text: ChatGPT, Claude, Gemini — drafting, summarising, coding, Q&A.;
- Images: DALL-E, Midjourney, Stable Diffusion — design, marketing, concept art.
- Audio: Suno, ElevenLabs — music generation, voice cloning, dubbing.
- Video: Sora, Runway — short-form video from text prompts.
- Code: GitHub Copilot, Cursor — autocomplete, refactoring, test generation.

Key Takeaways

- Discriminative AI classifies; Generative AI creates.
- Three shifts drove the revolution: scale, multimodality, and accessibility.
- Gen AI now spans text, images, audio, video, and code.

Quick-Revision Questions

- What is the fundamental difference between discriminative and generative AI?
- Name the three core shifts of the Generative AI revolution.
- Give one real-world example for each modality (text, image, audio, code).